Sampling, not learning: how compute, schedule, and decoding govern discrete diffusion on formal languages

Draft — oracle sampling-dynamics study

Abstract. We study discrete-diffusion generation of structured sequences under an **optimal denoiser**: an analytic grammar oracle that returns the exact valid-token distribution at every step. Holding the model perfect removes representation learning from the picture, so whatever variation remains is a property of **inference** alone. Across six formal languages we show that the accuracy a sampler reaches — and the accuracy–diversity trade-off it pays — is governed by three independent levers: **compute** (number of denoising steps), the **noise schedule**, and the **decoding rule** (greedy vs. categorical). Greedy buys accuracy at tiny compute but collapses to a single string; categorical keeps diversity but pays for accuracy in steps; a tuned Gaussian schedule dominates the uniform one at matched compute; and an adaptive (entropy-bounded) sampler reaches the accuracy frontier at a fraction of the steps. Because the denoiser is exact, these effects cannot be attributed to a model that failed to learn — they are intrinsic to diffusion sampling.

1 Setup and the oracle

We evaluate six formal languages, each equipped with an exact membership test that splits into two rules (e.g. an alphabet/format rule and a counting/parity rule); the reported metric is **both-rules accuracy**, the fraction of generated strings that satisfy both. The six grammars are ba^{N} (parity), $\mathrm{bba}^{\mathrm{N}}$, $\mathrm{a}^{\mathrm{N}}\mathrm{b}^{\mathrm{N}}$ and $\mathrm{a}^{\mathrm{N}}\mathrm{b}^{\mathrm{N}}\mathrm{c}^{\mathrm{N}}$ (counting), and Dyck-2 in a nested and a flat variant.

The central methodological move is the **oracle denoiser**. For each grammar we compute, analytically, the exact distribution over valid tokens at every masked position given the currently revealed context (`src/oracle/grammar_oracles.py`); oracle correctness is pinned by `tests/test_grammar_oracles.py` (38 passing tests). The oracle is the theoretical ceiling of any learned denoiser, so a sweep run on top of it isolates the **inference dynamics** of discrete diffusion from the learning problem entirely.

The sweep (`results/combined_6_grammar.csv`, 1{,}236 rows) crosses four decoders — uniform, Gaussian (sweeping σ), entropy-bounded adaptive ("EB", sweeping γ), and an autoregressive (AR) baseline — with two samplers (greedy/argmax and categorical/sampling) over requested steps $T \in [1, 1024]$. The compute axis throughout is **realised mean steps** (number of function evaluations, NFE), which is comparable across decoders. Two sequence-length regimes are baked in and annotated per panel rather than hidden: the single-string grammars (ba^{N}, $\mathrm{bba}^{\mathrm{N}}$, $\mathrm{a}^{\mathrm{N}}\mathrm{b}^{\mathrm{N}}$, $\mathrm{a}^{\mathrm{N}}\mathrm{b}^{\mathrm{N}}\mathrm{c}^{\mathrm{N}}$) are swept at $L = 128$, the two Dyck-2 grammars at $L = 32$. Per grammar we plot one diversity metric (uniqueness for ba^{N} / Dyck, n-coverage / n-entropy / $n\{,\}m$-joint-coverage for the counting grammars); cells flagged `n_correct_too_low` — too few valid samples to estimate diversity reliably (196 of 1{,}236 rows) — are dropped from the diversity figures by construction.

All figures and the companion `paper_figures/outlier_analysis.md` are regenerated from the single CSV by `paper_figures/make_paper_figures.py`, so every number quoted below traces to a plotted point.

2 The role of compute

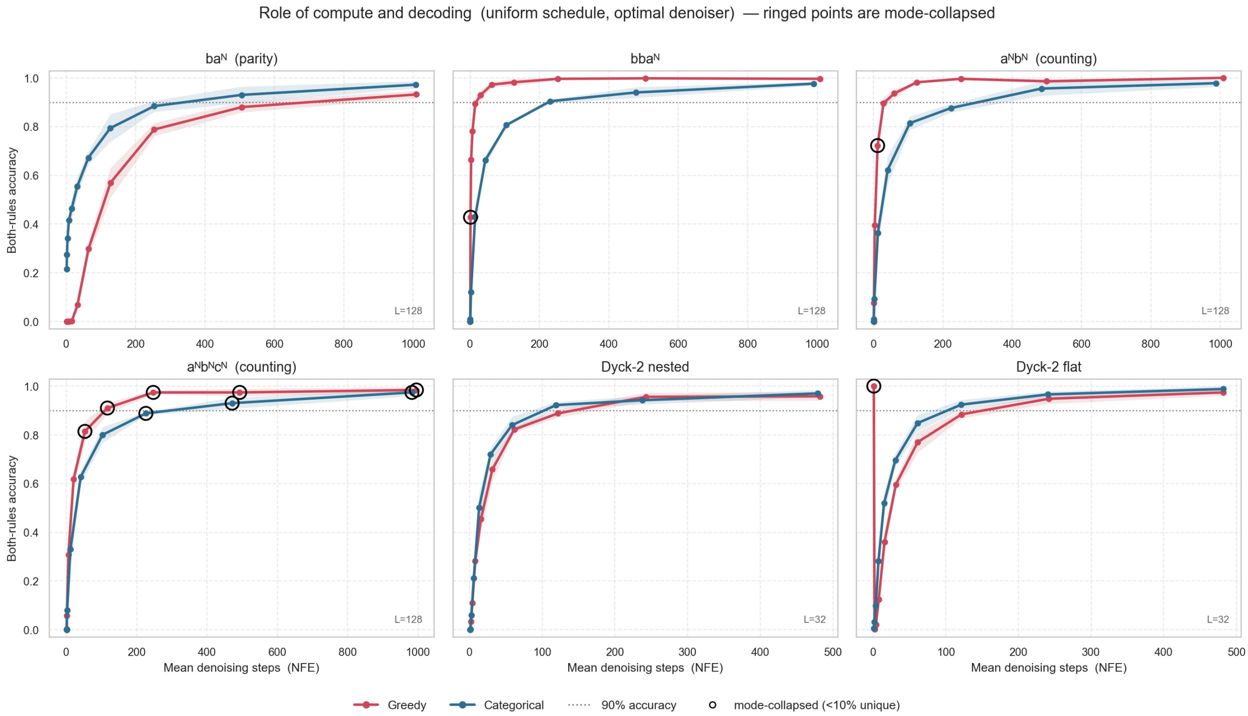

Figure 1: Both-rules accuracy vs. realised mean denoising steps (NFE) under the uniform schedule with the optimal denoiser. Greedy (red) vs. categorical (blue); ribbons are ± one standard deviation across repetitions; the dotted line marks 90% accuracy. *Ringed points are mode-collapsed* (uniqueness < 10%): a high-accuracy ring is a single repeated valid string, not skill. Each panel annotates its sequence length L.

The two samplers behave qualitatively differently as compute grows (Figure 1). **Categorical accuracy rises monotonically with steps** and plateaus just *below* 1.0 even at the largest budget (here extended to ~ 1000 mean steps): the grammars settle in a narrow band — a^{N}b^N → 0.98, a^{N}b^{N}c^N → 0.97, ba^N → 0.97, Dyck-2 flat → 0.99 — because each undecided position is sampled afresh and a perfectly satisfied global constraint requires every draw to land right. **Greedy, by contrast, saturates almost immediately** — but the early saturation is often illusory. Dyck-2 flat reaches 1.000 at $T = 1$, yet its uniqueness is just 0.002: greedy argmaxes the fully-masked marginal in one shot, and when the modal per-position string happens to be grammatical the cell scores 1.0 while emitting the *same* string every time (the ringed points). This is why accuracy at low T is only meaningful read together with diversity.

More compute is not always better. On the best-per-budget envelope, 12 decoder × sampler lines peak at intermediate compute and then decline — almost all of them under the Gaussian schedule, where an over-wide σ at the largest T hurts (e.g. Dyck-2 flat greedy peaks at 1.000 but ends at 0.898; ba^N Gaussian categorical peaks at 0.996, ends at 0.726). The effect is enumerated in `outlier_analysis.md` §6 and revisited in Section 7.

3 Schedule and adaptive sampling

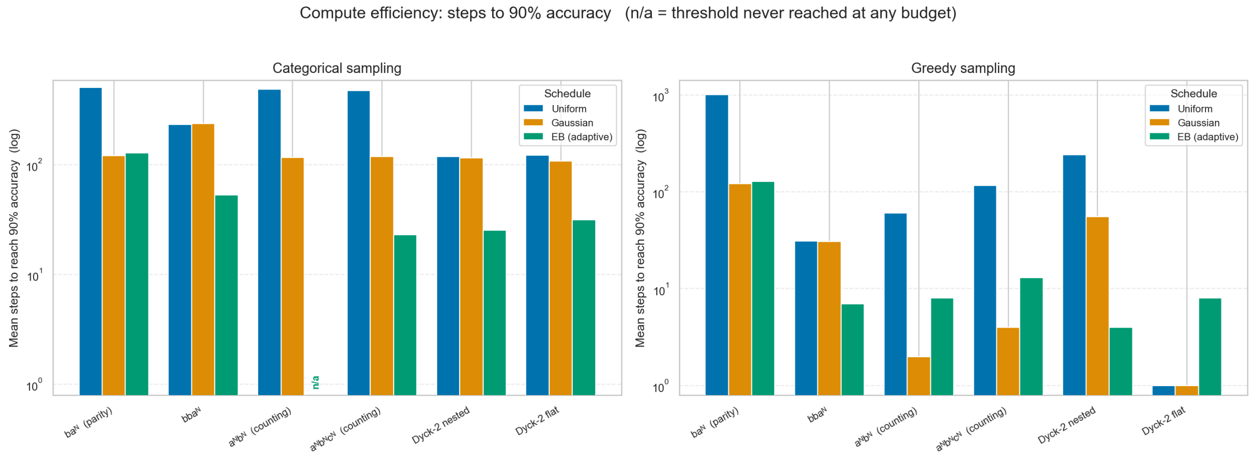

Figure 2: Compute efficiency: mean denoising steps needed to first reach 90% accuracy (log scale), per grammar and schedule, split by sampler. Lower is better; “n/a” marks a schedule that never reaches 90% at any budget. The adaptive EB sampler (green) reaches the bar at far fewer steps than the fixed uniform/Gaussian schedules, especially under greedy decoding.

Holding accuracy fixed at the 90% bar and reading off the *cost* in steps (Figure 2) separates the schedules cleanly. **A tuned Gaussian schedule dominates the uniform one at matched compute**: on a^Nb^N, Gaussian-greedy with $\sigma = 2$ reaches 1.000 in about 4 steps where uniform-greedy needs ~ 60; and Gaussian-categorical reaches 90% in ~ 117 steps versus ~ 484 for uniform — roughly a 4× saving. **The adaptive EB sampler is the Pareto-efficient operating point**: under greedy it hits 90% in 4 (Dyck-2 nested), 8 (a^Nb^N, Dyck-2 flat) and 13 ($a^Nb^Nc^N$) mean steps — one to two orders of magnitude below the fixed schedules, which now range from tens to ~ 1000 steps.

The efficiency win is *not* unconditional, and we flag it rather than overclaim. Under **categorical** sampling EB underperforms on the counting grammars — a^Nb^N EB-categorical never reaches the 90% bar at all (it tops out at 0.620, vs. 1.000 for EB-greedy) — because entropy-bounded stopping halts while the count is still uncertain under stochastic draws. EB should therefore be presented as Pareto-efficient for greedy decoding, not as a uniformly dominant sampler.

4 The accuracy–diversity trade-off

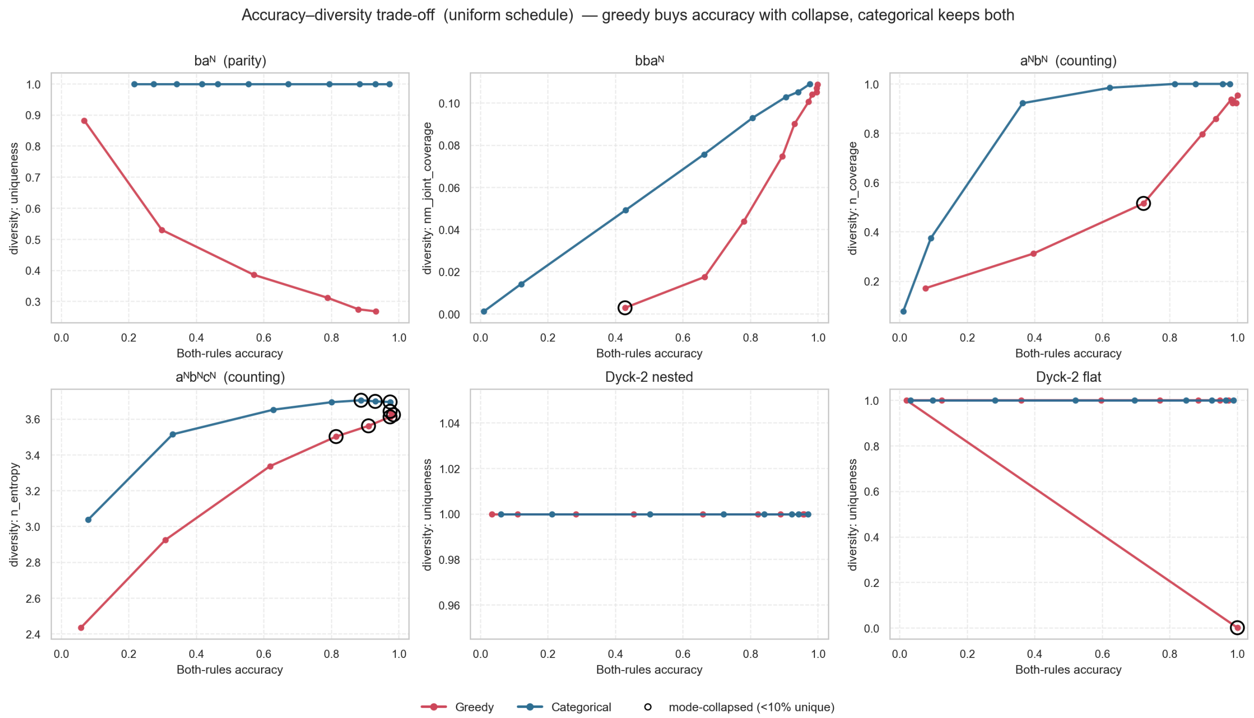

Figure 3: Per-grammar diversity vs. both-rules accuracy under the uniform schedule, points ordered along each line by compute. Greedy (red) hugs the bottom — accuracy bought with collapse; categorical (blue) climbs to the top-right — accurate *and* diverse. Ringed points are mode-collapsed (uniqueness $< 10\%$).

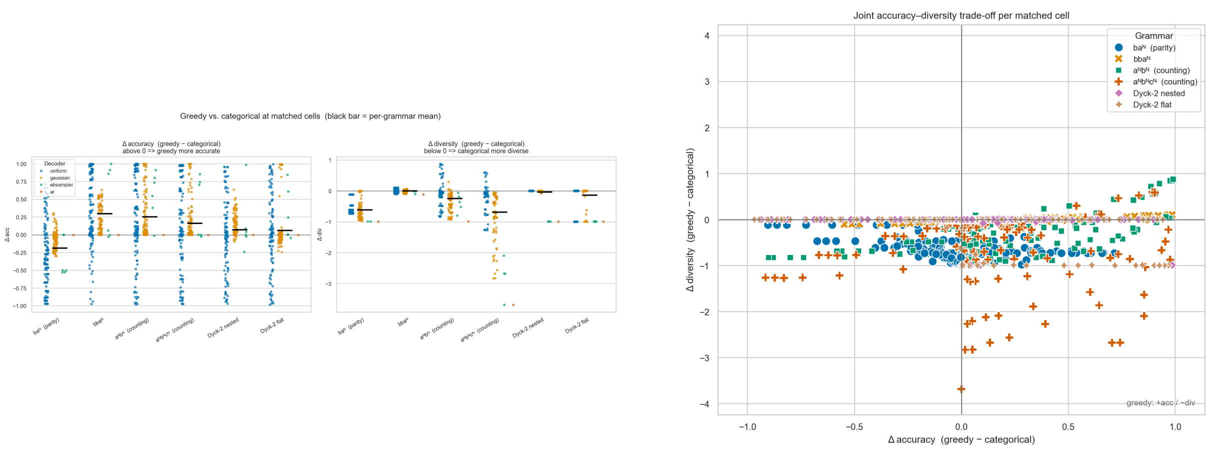

Figure 4: Greedy minus categorical at *matched* cells (same grammar, schedule, hyperparameter, T). Left: paired Δ-accuracy and Δ-diversity strips (black bar = per-grammar mean). Right: the joint Δ-accuracy / Δ-diversity quadrant — the cloud sits below zero on the diversity axis (categorical more diverse) and spreads both ways on accuracy, the signature of a trade-off rather than a free lunch.

The trade-off is the through-line of the study (Figure 3, Figure 4). **Greedy buys accuracy with collapse** — its uniqueness sits near the floor (0.01) across grammars, so the high-accuracy greedy points are concentrated point masses. **Categorical buys diversity with compute** — uniqueness climbs toward 1.0 for ba^N and Dyck as accuracy rises, at the cost of the many steps documented in Section 2. In the matched quadrant the point cloud lies almost entirely in the lower half-plane (categorical strictly more diverse) while straddling zero on accuracy: there is no setting that is simultaneously more accurate *and* more diverse.

Crucially, **the AR baseline shows the same trade-off**, which argues it is a property of the sampling regime and not of diffusion specifically. AR reaches 1.000 accuracy on every grammar under both samplers — but greedy-AR has uniqueness 0.010 everywhere (one memorised modal string), whereas categorical-AR recovers full diversity (uniqueness 1.0 on ba^N and both Dyck variants). Greedy-vs-categorical reproduces the accuracy/diversity tension whether the backbone is diffusion or autoregressive.

5 Mechanism: oracle-logit monotonicity

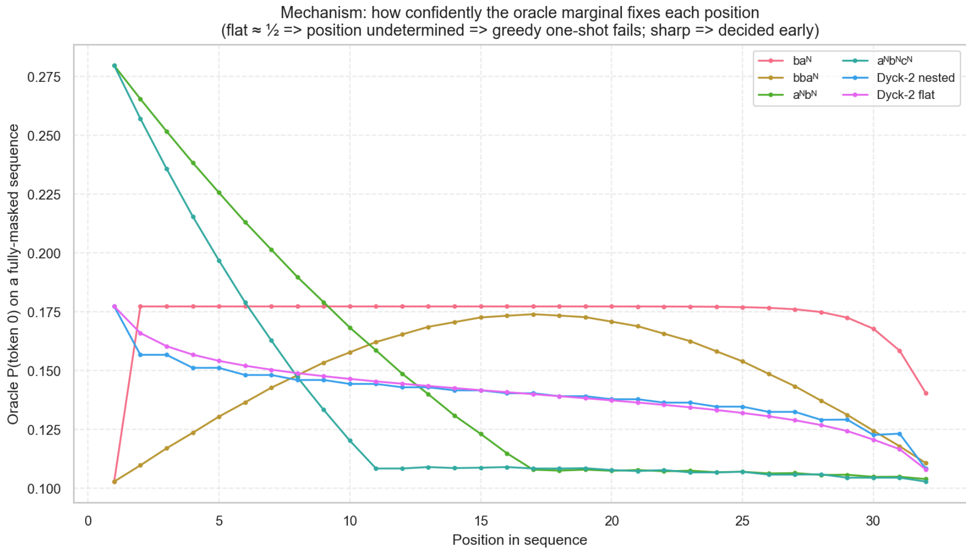

Figure 5: Mechanism. Oracle marginal probability of the first content symbol at each position on a fully-masked sequence. A flat ~ 1/ 2 profile means the position is *undetermined* by the marginal alone (so one-shot greedy must guess and fails); a sharp profile means the position is *decided early* (so greedy succeeds at low T).

The positional logit profiles explain *why* the samplers diverge (Figure 5). For grammars whose per-position oracle marginal is sharp, the correct symbol is fixed by context almost immediately, so greedy commits correctly in one or a few steps — matching its early saturation in Figure 1. For grammars governed by a *global* constraint, interior positions sit near ~ 1/ 2 under the marginal because either symbol can be valid depending on the rest of the string; the position is only resolved once enough neighbours are committed and the marginal re-conditions. This is exactly the regime where greedy's one-shot argmax picks a globally inconsistent pattern while categorical's fair draw at least matches the chance rate — the bridge from the mechanism to the sampler behaviour of Section 2 and Section 4.

5.1 The ba^N parity anomaly

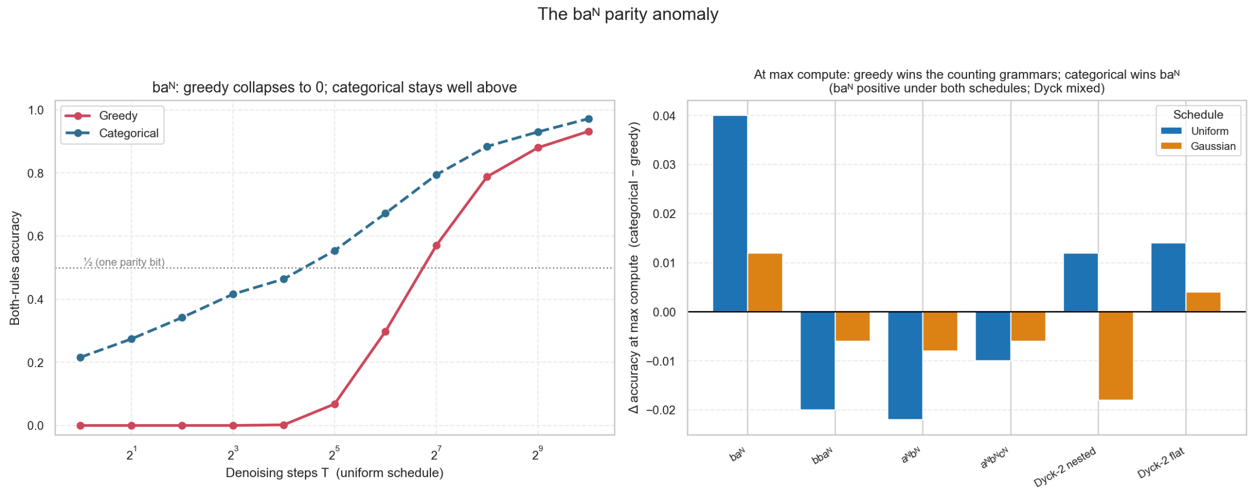

Figure 6: The ba^N parity outlier. Left: under the uniform schedule ba^N greedy is pinned at 0 for $T \leq 8$ while categorical is already nonzero at $T = 1$ (~ 0.22) and climbs steadily, crossing the ½ parity-bit reference near $T \approx 8$. Right: categorical minus greedy accuracy *at maximum compute* per schedule — comparing at the largest budget sidesteps greedy's low-T collapse, since there greedy is fully iterative and diverse.

ba^N ("starts with b, even number of a's") is the cleanest single piece of evidence that the failure is a *sampling* phenomenon, not a denoiser-capacity one — the denoiser is exact (Figure 6). The even-count rule is a global parity constraint, so the oracle's per-position marginal is about 1/ 2 at interior slots. Uniform greedy at $T \leq 8$ therefore scores 0.000 (it commits to one fixed pattern whose parity is wrong with near-certainty), while uniform categorical is already nonzero at $T = 1$ (0.216, against greedy's exact 0) and climbs steadily as conditioning kicks in — the residual gap below ½ at $T = 1$ is the joint format rule, which a single independent draw rarely satisfies as well.

We are careful not to overclaim ba^N as unique. Comparing at *maximum compute* (the degeneracy-free regime), categorical beats greedy on ba^N and both Dyck variants, with ba^N the widest margin (0.972 vs. 0.932 under uniform) and the only case positive under *both* schedules; greedy still wins all three counting grammars. The headline is that ba^N is the *strongest*, not the sole, instance of marginal-vs-joint sampling failure.

6 Comparing the three decoders

Sections Section 2 and Section 4 fixed the schedule to uniform so that the sampler effect could be read cleanly. Here we lift that restriction and put all three diffusion decoders — uniform, Gaussian, and adaptive EB — on the same axes, so the compute and trade-off stories of those sections generalise rather than describe one schedule. Each line is the best-per-budget envelope (best σ / γ chosen at each compute budget); the AR baseline is a single point, not a sweep, and is omitted.

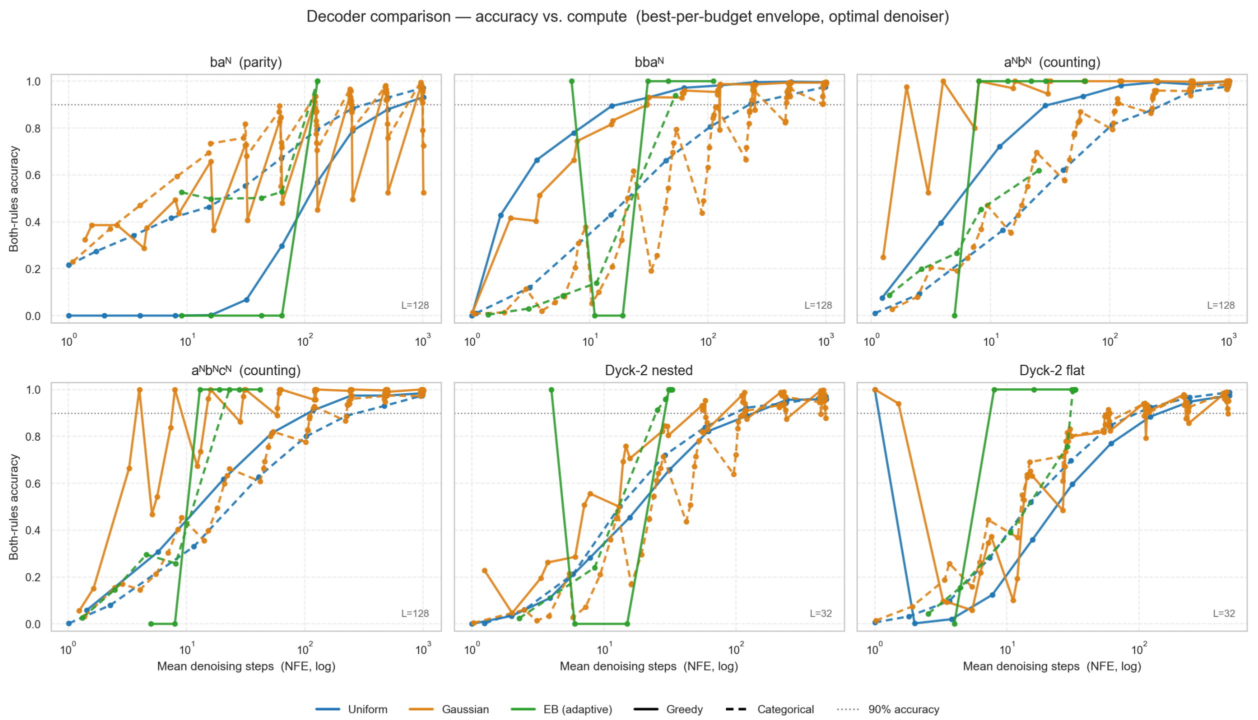

Figure 7: Accuracy vs. compute for all three decoders (colour) under both samplers (solid = greedy, dashed = categorical), best-per-budget envelope, log NFE. EB (green) reaches high accuracy at the fewest steps; uniform (blue) is the slowest climber; Gaussian (orange) sits between, and its envelope is jagged because the best σ changes with budget.

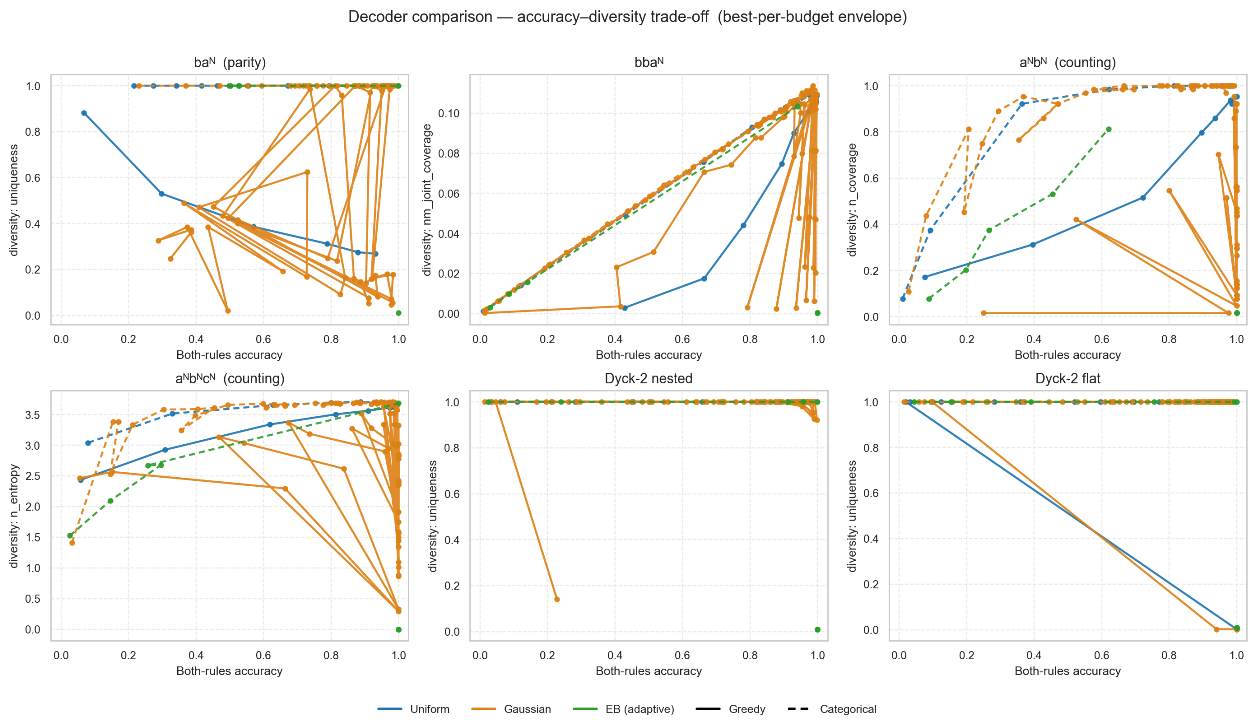

Figure 8: Accuracy–diversity trade-off for all three decoders (same encoding as Figure 7), points ordered by compute. The decoder shifts **where** on the trade-off a sampler operates but does not remove the greedy-vs-categorical split of Section 4.

Three contrasts stand out (Figure 7, Figure 8). **The decoder sets compute efficiency, not the ceiling.** All three schedules ultimately reach similar accuracy on most grammars; what differs is how many steps it costs — EB reaches the 90% bar one to two orders of magnitude sooner than uniform

(Section 3). **Gaussian is the consistent middle.** It dominates uniform almost everywhere (quantified in Section 7) while not matching EB's adaptive step savings. **The sampler split survives the decoder.** Under every decoder, greedy still hugs the low-diversity region and categorical still climbs to the accurate-and-diverse corner (Figure 8): the accuracy–diversity tension of Section 4 is a property of the decoding rule, robust to the choice of schedule.

7 The Gaussian schedule

The remaining sections fixed or pooled the schedule; we now study the Gaussian schedule on its own terms, because it is the one schedule with a tunable shape parameter and it turns out to interpolate between two familiar regimes.

7.1 Definition

Let L be the sequence length and $t \in [0, 1]$ a continuous timestep ($t = 0$ clean, $t = 1$ fully masked). For a width parameter $\sigma > 0$ define the sweep width and a mean that sweeps monotonically from right to left across the sequence,

$$W = L + 4\sigma, \quad \mu(t) = (1 - t)W - 2\sigma,$$

and set the per-position masking probability to the Gaussian CDF Φ evaluated at the signed distance from that mean,

$$p_{\mathrm{mask},i}(t) = \Phi\left(\frac{i - \mu(t)}{\sigma}\right), \quad \alpha_i(t) = 1 - p_{\mathrm{mask},i}(t).$$

At $t = 0$ the mean sits far to the right ($\mu = L + 2\sigma$) so $p_{\mathrm{mask},i} \approx 0$; at $t = 1$ it sits far to the left ($\mu = -2\sigma$) so $p_{\mathrm{mask},i} \approx 1$. Training uses the standard masked-diffusion ELBO weight $\left(\partial p_{\mathrm{mask},i} \,/\, \partial t\right) \,/\, p_{\mathrm{mask},i}(t)$ (here position-dependent rather than uniform), with the EOS token upweighted by $\lambda > 1$. The implementation is `src/schedules/gaussian_schedule.py` and matches these equations exactly.

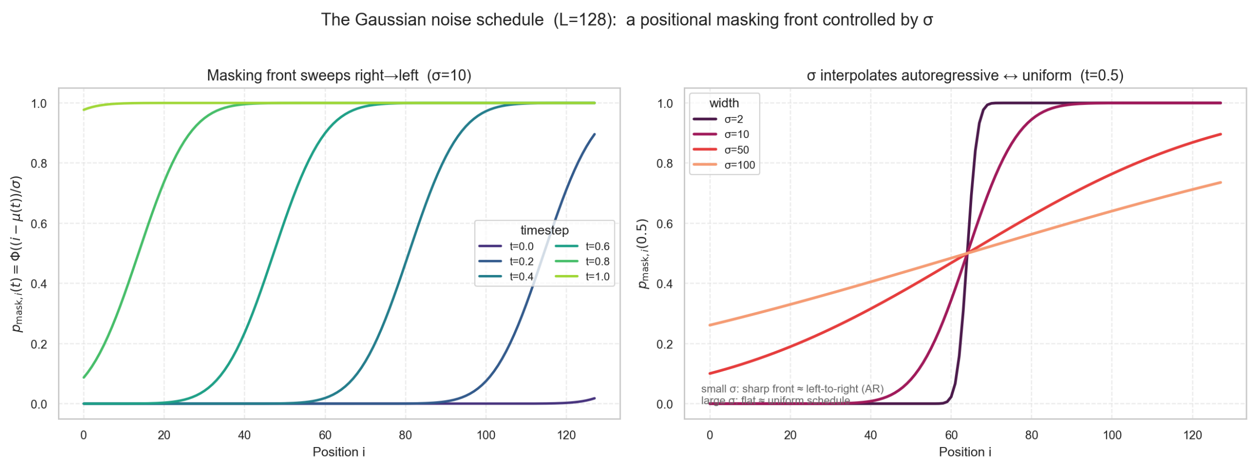

Figure 9: The Gaussian schedule, regenerated from the source implementation ($L = 128$). Left: the masking front sweeps right→ left as t grows, so suffix positions are revealed only after their prefix context exists. Right: at fixed t, σ morphs the front from a sharp left-to-right step (small σ, essentially autoregressive) to a flat profile (large σ, essentially the uniform schedule).

The shape parameter is the whole point (Figure 9). At **small** σ the active window — the slanted part of the CDF where $0 < p_{\mathrm{mask},i} < 1$ — is only a few tokens wide, so positions are committed left-to-right with full prefix context: the schedule **approximates autoregressive generation**. As σ grows the front flattens and approaches a constant, **recovering the uniform schedule** (and its pathology of revealing suffix tokens while the prefix is still masked). The uniform schedule is thus the large-σ limit of this family, and σ is a continuous knob between autoregressive and uniform sampling. (The motivating source note observed, on a **trained** RPE transformer for $a^N b^N$, that the uniform schedule capped at

0.955 accuracy while a tuned Gaussian reached 1.0 with a ~ $5\times$ step saving; trained models are out of scope here, but the same schedule is what we now probe with the optimal denoiser.)

7.2 Effect of T and σ on accuracy

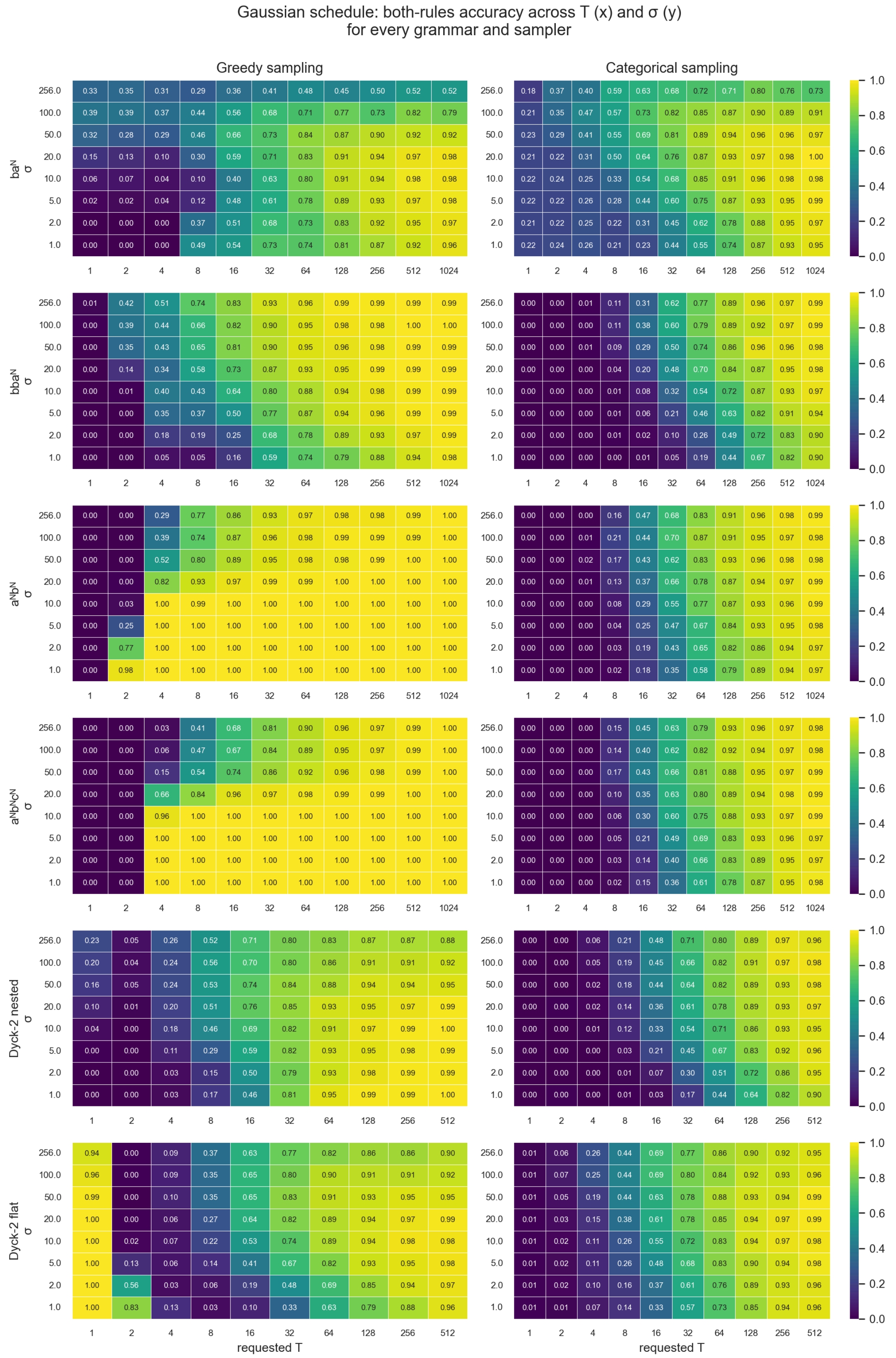

Figure 10: Both-rules accuracy across requested steps T (x) and width σ (y) for every grammar (rows) and sampler (columns), Gaussian schedule, optimal denoiser. Accuracy rises rightward with compute everywhere; the σ direction behaves **oppositely** for the two samplers.

Reading Figure 10 across the two axes exposes a sampler-dependent σ effect that a single-schedule view hides. **Compute (T) helps monotonically** for both samplers, as in Section 2. **The best σ, however, depends on the sampler. Greedy** thrives on the sharp, autoregressive-like front of a **small** σ: the counting grammars reach 1.000 almost for free there (a^Nb^N at σ= 2 hits 1.000 in just 4 steps; $a^Nb^Nc^N$ at σ= 1, T{=}4), and a large σ — approaching the uniform limit — hurts. The penalty is masked on the counting grammars once compute is ample (with T{=}1024 greedy reaches 1.000 at every σ), but it is unmistakable on ba^N, whose greedy accuracy at T{=}1024 falls from 0.98 at σ= 10–20 to 0.79 at σ= 100 and 0.52 at σ= 256. **Categorical** instead prefers a **moderate** σ: ba^N categorical at T{=}1024 rises from 0.95 at σ= 1 to a peak of 1.00 at σ= 20, then falls to 0.73 at σ= 256, because a wider front gives the stochastic sampler re-conditioning opportunities but the uniform limit reintroduces context-free unmasking. ba^N shows the coupling most sharply — parity needs **some** spread but is destroyed as σ → the uniform limit.

7.3 Gaussian vs. uniform across all metrics

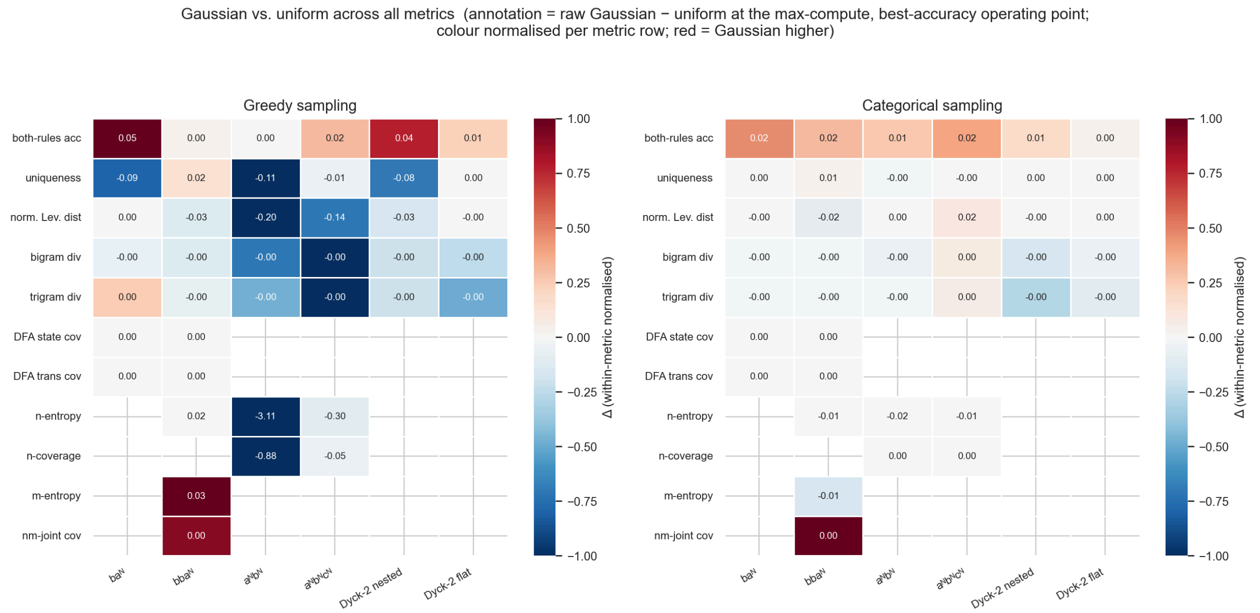

Figure 11: Gaussian minus uniform across every recorded metric (rows) per grammar (columns) at the max-compute, best-accuracy operating point, split by sampler. Annotations are the raw difference; colour is normalised within each metric row so bounded metrics are not swamped by the unbounded entropy rows. Red = Gaussian higher. Blank = metric undefined for that grammar or gated by the diversity-reliability flag.

At the degeneracy-free max-compute operating point (Figure 11), **Gaussian dominates uniform on accuracy** almost everywhere; with both schedules now run to ~ 1000 steps the margins are smaller than at lower budgets but the sign is consistent. Under greedy Gaussian wins ba^N by +0.052, Dyck-2 nested by +0.040, $a^Nb^Nc^N$ by +0.016 and Dyck-2 flat by +0.012, with bba^N +0.002 and a^Nb^N a tie; under categorical it wins **all six** grammars (ba^N +0.024, $a^Nb^Nc^N$ +0.020, bba^N +0.016, a^Nb^N +0.014, Dyck-2 nested +0.010, Dyck-2 flat +0.002). **On diversity the picture is mixed and sampler-specific.** For categorical the diversity metrics barely move (Gaussian keeps uniform's diversity while adding accuracy — a near-free win). For greedy the accuracy gains come partly at a **diversity cost**: Gaussian-greedy lowers uniqueness by 0.089 on ba^N and by 0.112 on a^Nb^N, consistent with its more autoregressive, more committed decoding. The practical reading: **a tuned Gaussian schedule is a strict improvement over uniform for categorical sampling, and an accuracy-for-diversity trade for greedy** — never worse on the headline accuracy metric.

8 Conclusions

With the denoiser held optimal, three inference levers independently govern discrete-diffusion generation of structured sequences: **compute** sets how far categorical accuracy climbs (and greedy saturates almost free); the **noise schedule** sets the cost of a given accuracy (tuned Gaussian dominates uniform; adaptive EB reaches the frontier at a fraction of the steps under greedy); and the **decoding rule** sets where on the accuracy–diversity trade-off you land (greedy: accurate but collapsed; categorical: diverse but compute-hungry). The trade-off persists even with a perfect denoiser and reappears in the AR baseline, so it is intrinsic to the sampling regime rather than to any failure of learning. Putting all three decoders on the same axes (Section 6) confirms the picture generalises: the decoder changes compute efficiency and the operating point, not the sampler split.

The Gaussian schedule deserves a separate note (Section 7). Its width σ is a continuous knob between an autoregressive-like sharp front (small σ) and the uniform schedule (large σ), and the best σ is **sampler-dependent**: greedy wants a small, AR-like σ, categorical a wider one. Tuned, it beats uniform on accuracy on almost every grammar — strictly so for categorical, and as an accuracy-for-diversity trade for greedy.

Practical guidance: prefer the adaptive EB sampler with a tuned Gaussian schedule under greedy decoding for accuracy at low compute; switch to categorical when output diversity is required and budget allows. **Future work:** place trained models against this oracle ceiling — how closely real denoisers approach the frontier, and whether rule extrapolation and length generalisation survive the same sampling pressures — is the natural next step beyond the scope of this study.